

DATABASE MANAGEMENT SYSTEM

Seller and Inventory Management System

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Task	Task 3
Subject	Database Management System
Task Name	Seller and Inventory Management System

Description

Create Seller and Inventory tables. Establish relationships between sellers, products, and stock. Maintain seller product information. Track available and unavailable products. Generate inventory status reports.

1. Project Objective

The objective is to design a relational database for managing sellers, products, and inventory stock. The system maintains seller-product information, records available stock, identifies unavailable products, and supports inventory status reporting.

The database uses primary keys and foreign keys to maintain relationships between sellers, products, and inventory records.

2. Database and Seller Table

A database named SellerInventoryDB can be used to organize the seller and inventory information. The Seller table stores seller identification and contact information.

```
CREATE DATABASE SellerInventoryDB;
USE SellerInventoryDB;

CREATE TABLE Seller (
    seller_id INT PRIMARY KEY,
    seller_name VARCHAR(100) NOT NULL,
    email VARCHAR(100),
    phone VARCHAR(15)
);
```

Each seller is uniquely identified by seller_id. The NOT NULL constraint ensures that a seller name is provided.

3. Product and Inventory Tables

The Product table stores product information, while the Inventory table connects products with sellers and records the available stock quantity and status.

```
CREATE TABLE Product (  
    product_id INT PRIMARY KEY,  
    product_name VARCHAR(100) NOT NULL,  
    price DECIMAL(10,2)  
);  
  
CREATE TABLE Inventory (  
    inventory_id INT PRIMARY KEY,  
    seller_id INT,  
    product_id INT,  
    stock_quantity INT,  
    stock_status VARCHAR(20),  
    FOREIGN KEY (seller_id)  
        REFERENCES Seller(seller_id),  
    FOREIGN KEY (product_id)  
        REFERENCES Product(product_id)  
);
```

The foreign keys establish the relationship between sellers, products, and their inventory records.

4. Sample Seller and Product Data

Sample records can be inserted to test seller product management and inventory tracking.

```
INSERT INTO Seller
(seller_id, seller_name, email, phone)
VALUES
(1, 'ABC Electronics', 'abc@gmail.com', '9876500001'),
(2, 'Tech World', 'techworld@gmail.com', '9876500002'),
(3, 'Smart Mart', 'smartmart@gmail.com', '9876500003');
```

```
INSERT INTO Product
(product_id, product_name, price)
VALUES
(101, 'Laptop', 55000.00),
(102, 'Mobile Phone', 25000.00),
(103, 'Keyboard', 1500.00),
(104, 'Mouse', 800.00),
(105, 'Headphones', 2000.00);
```

5. Inventory Records

Inventory records associate a seller with a product and store the current quantity. The stock status can be maintained as Available or Unavailable.

```
INSERT INTO Inventory
(inventory_id, seller_id, product_id, stock_quantity, stock_status)
VALUES
(1, 1, 101, 10, 'Available'),
(2, 1, 103, 25, 'Available'),
(3, 2, 102, 0, 'Unavailable'),
(4, 2, 104, 15, 'Available'),
(5, 3, 105, 0, 'Unavailable');
```

A quantity of zero can be used to represent products that are currently unavailable, while positive quantities represent available stock.

6. Maintain Seller Product Information

Seller-product information can be retrieved through JOIN operations. This allows the system to display which seller supplies a particular product and the associated stock quantity.

```
SELECT
    Seller.seller_name,
    Product.product_name,
    Product.price,
    Inventory.stock_quantity,
    Inventory.stock_status
FROM Seller
JOIN Inventory
    ON Seller.seller_id = Inventory.seller_id
JOIN Product
    ON Inventory.product_id = Product.product_id
ORDER BY Seller.seller_name;
```

This query provides a combined view of seller, product, price, stock quantity, and availability status.

7. Track Available and Unavailable Products

Available products can be identified by checking positive stock or an Available status. Unavailable products can be identified using a zero quantity or Unavailable status.

```
-- Available products
SELECT
    Product.product_name,
    Seller.seller_name,
    Inventory.stock_quantity
FROM Product
JOIN Inventory
    ON Product.product_id = Inventory.product_id
JOIN Seller
    ON Inventory.seller_id = Seller.seller_id
WHERE Inventory.stock_quantity > 0;

-- Unavailable products
SELECT
    Product.product_name,
    Seller.seller_name,
    Inventory.stock_quantity
FROM Product
JOIN Inventory
    ON Product.product_id = Inventory.product_id
JOIN Seller
    ON Inventory.seller_id = Seller.seller_id
WHERE Inventory.stock_quantity = 0;
```


8. Inventory Status Reports

An inventory report summarizes the current stock state of products and sellers.

```
SELECT
    Seller.seller_name,
    COUNT(Inventory.product_id) AS total_products,
    SUM(CASE
        WHEN Inventory.stock_quantity > 0 THEN 1
        ELSE 0
    END) AS available_products,
    SUM(CASE
        WHEN Inventory.stock_quantity = 0 THEN 1
        ELSE 0
    END) AS unavailable_products,
    SUM(Inventory.stock_quantity) AS total_stock
FROM Seller
LEFT JOIN Inventory
    ON Seller.seller_id = Inventory.seller_id
GROUP BY Seller.seller_id, Seller.seller_name;
```

The report gives a seller-wise summary of total products, available products, unavailable products, and total stock quantity.

9. Inventory Update Operations

Stock quantities and availability can be updated whenever products are received, sold, or become unavailable.

```
UPDATE Inventory
SET stock_quantity = 20,
    stock_status = 'Available'
WHERE inventory_id = 3;
```

```
UPDATE Inventory
SET stock_quantity = 0,
    stock_status = 'Unavailable'
WHERE inventory_id = 4;
```

These operations demonstrate how inventory status can be maintained as stock levels change.

10. Database Relationships and Expected Output

Table	Key	Relationship / Purpose
Seller	seller_id	Identifies each seller
Product	product_id	Identifies each product
Inventory	inventory_id	Stores seller-product stock information
Inventory → Seller	seller_id	Links inventory records to sellers
Inventory → Product	product_id	Links inventory records to products

Expected Results:

- Seller details can be stored and retrieved.
- Products can be associated with individual sellers.
- Stock quantities can be maintained.
- Available and unavailable products can be identified.
- Seller-wise inventory status reports can be generated.

Conclusion

The Seller and Inventory Management System demonstrates a relational DBMS solution for maintaining seller information, product details, and stock status. Primary and foreign keys establish the relationships, while SQL JOIN, filtering, aggregation, and UPDATE operations support inventory management and reporting.